CO2 - Assessment Tool 1 -database and connectivity

CODE: DSA0506-QUERRY PROCESSSING FOR DATASCIENCE

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1. **1. Employee and Department Management System (10 Marks)**

A company maintains employee details and department information in separate PostgreSQL tables. Design **Department** and **Employee** tables using appropriate primary and foreign keys. Write a Python program to connect to the PostgreSQL database, create the tables, insert sample records, and display employee names along with their respective department names using an **SQL JOIN** query.

Sol;

```
conn = sqlite3.connect(':memory:')

cur = conn.cursor()

create_department_table_sql = """

CREATE TABLE Department(

    Department_ID INTEGER PRIMARY KEY AUTOINCREMENT,

    Department_Name VARCHAR(50)

);

"""cur.execute(create_department_table_sql)

create_employee_table_sql = """

CREATE TABLE Employee(

    Employee_ID INTEGER PRIMARY KEY AUTOINCREMENT,

    Employee_Name VARCHAR(50),

    Salary NUMERIC(10,2),

    Department_ID INT,
```

```
    FOREIGN KEY (Department_ID)
    REFERENCES Department(Department_ID)
);
"""cur.execute(create_employee_table_sql)
conn.commit()
cur.close()
conn.close()
print("Tables 'Department' and 'Employee' created successfully in an in-memory SQLite database.")
```

```
Tables 'Department' and 'Employee' created successfully in an in-memory SQLite database.
```

```
import sqlite3
in all relevant cells.
create_department_table_sql = """
CREATE TABLE Department(
    Department_ID INTEGER PRIMARY KEY AUTOINCREMENT,
    Department_Name VARCHAR(50)
);
"""cur.execute(create_department_table_sql)
create_employee_table_sql = """
CREATE TABLE Employee(
    Employee_ID INTEGER PRIMARY KEY AUTOINCREMENT,
    Employee_Name VARCHAR(50),
    Salary NUMERIC(10,2),
    Department_ID INT,
```

```
    FOREIGN KEY (Department_ID)
    REFERENCES Department(Department_ID)
);
"""cur.execute(create_employee_table_sql)
insert_department_sql = """
INSERT INTO Department(Department_Name)
VALUES
('HR'),
('IT'),
('Finance');
"""cur.execute(insert_department_sql)
insert_employee_sql = """
INSERT INTO Employee(Employee_Name,Salary,Department_ID)
VALUES
('Rahul',50000,1),
('Priya',60000,2),
('Anil',55000,2),
('Sneha',65000,3);
"""
cur.execute(insert_employee_sql)
conn.commit()
cur.close()
conn.close()
print("Data inserted successfully into 'Department' and 'Employee' tables.")
```

```
import sqlite3

conn = sqlite3.connect(':memory:')

cur = conn.cursor()

create_department_table_sql = """

CREATE TABLE Department(

    Department_ID INTEGER PRIMARY KEY AUTOINCREMENT,

    Department_Name VARCHAR(50)

);

"""

cur.execute(create_department_table_sql)

create_employee_table_sql = """

CREATE TABLE Employee(

    Employee_ID INTEGER PRIMARY KEY AUTOINCREMENT,

    Employee_Name VARCHAR(50),

    Salary NUMERIC(10,2),

    Department_ID INT,

    FOREIGN KEY (Department_ID)

    REFERENCES Department(Department_ID)

);

"""

cur.execute(create_employee_table_sql)

insert_department_sql = """

INSERT INTO Department(Department_Name)

VALUES
```

```
('HR'),
('IT'),
('Finance');
"""
cur.execute(insert_department_sql)
insert_employee_sql = """
INSERT INTO Employee(Employee_Name,Salary,Department_ID)
VALUES
('Rahul',50000,1),
('Priya',60000,2),
('Anil',55000,2),
('Sneha',65000,3);
"""cur.execute(insert_employee_sql)
query = """
SELECT Employee.Employee_Name,
       Department.Department_Name
FROM Employee
JOIN Department
ON Employee.Department_ID = Department.Department_ID;
"""cur.execute(query)
rows = cur.fetchall()
print("Employee Name\tDepartment")
for row in rows:
    print(row[0], "\t", row[1])
```

```
conn.commit()

cur.close()

conn.close()

print("\nQuery executed successfully using SQLite.")
```

2. Hotel Room Reservation System (10 Marks)

A hotel maintains room information and reservation details in an SQLite database. Design **Room** and **Reservation** tables using suitable primary and foreign keys. Write a Python program to connect to the database, create the tables, insert sample data, update the reservation status of a customer, and display the reservation details

```
import sqlite3

conn = sqlite3.connect("hotel.db")

cur = conn.cursor()

# Create Room table

cur.execute("""

CREATE TABLE IF NOT EXISTS Room(

Room_ID INTEGER PRIMARY KEY,

Room_Type TEXT,

Price INTEGER

)

""")

# Create Reservation table

cur.execute("""
```

```
CREATE TABLE IF NOT EXISTS Reservation(
Reservation_ID INTEGER PRIMARY KEY,
Customer_Name TEXT,
Room_ID INTEGER,
Status TEXT,
FOREIGN KEY(Room_ID) REFERENCES Room(Room_ID)
)
""")
# Insert sample rooms
cur.execute("INSERT INTO Room VALUES(1,'Single',1000)")
cur.execute("INSERT INTO Room VALUES(2,'Double',2000)")
cur.execute("INSERT INTO Room VALUES(3,'Suite',3000)")
# Insert reservation
cur.execute("INSERT INTO Reservation VALUES(101,'Rahul',1,'Booked')")
# Update reservation status
cur.execute("""
UPDATE Reservation
SET Status='Confirmed'
WHERE Reservation_ID=101
""")
# Display reservation details
cur.execute("""
SELECT Reservation.Customer_Name,
Room.Room_Type,
```

```
Reservation.Status

FROM Reservation

JOIN Room

ON Reservation.Room_ID=Room.Room_ID

""")

rows = cur.fetchall()

print("Customer\tRoom\tStatus")

for row in rows:

    print(row[0], "\t", row[1], "\t", row[2])

conn.commit()

conn.close()
```

```
Customer        Room    Status
Rahul    Single          Confirmed
```